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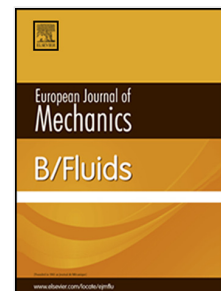
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A global approximation to the Green function for diffraction radiation of water waves

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Abstract

The Green function of the theory of diffraction radiation of time-harmonic (regular) waves by an offshore structure, or a ship at low speed, in deep water is considered. The Green function G and its gradient ∇G are expressed in the usual manner as the sum of three components that correspond to the fundamental free-space singularity, a non-oscillatory local flow, and waves. Simple approximations that only involve elementary continuous functions (algebraic, exponential, logarithmic) of real arguments are given for the local flow components in G and ∇G . These approximations are global approximations valid within the entire flow region, rather than within complementary contiguous regions as can be found in the literature. The analysis of the errors associated with the approximations to the local flow components given in the study shows that the approximations are sufficiently accurate for practical purposes. These global approximations provide a particularly simple and highly efficient way of numerically evaluating the Green function and its gradient for diffraction radiation of time-harmonic waves in deep water.

Keywords: regular water waves, diffraction radiation, Green function, local flow, global approximation

1. Introduction

Diffraction radiation of time-harmonic water waves by an offshore structure within the classical framework of linear potential flow theory and the Green function method is routinely used to predict added-mass and wave-damping coefficients, motions, and wave loads. Wave diffraction radiation by a ship that advances in ambient waves at low forward speed is also often analyzed using the zero-speed Green function at the encounter frequency. This Green function, which represents the velocity potential due to a pulsating source at a singular point under the free surface as is well known, is an essential element of the theory of wave diffraction radiation. Accordingly, the Green function has been studied in a broad literature, e.g. Havelock [1, 2], Thorne [3], Haskind [4], Wehausen [5], Kim [6], Noblesse [7], especially for the simplest case of deep water that is considered here.

The Green function G can be expressed as the sum of the fundamental free-space singularity and a flow component that accounts for free-surface effects. Moreover, this free-surface component is commonly decomposed into a wave component W that represents the waves radiated by the pulsating source, and a non-oscillatory local flow component L . This basic decomposition into a wave component and a local flow component is not unique. Indeed, three alternative decompositions and related single-integral representations of the Green function G are given in Noblesse [7].

Several alternative mathematical representations and approximations of G and ∇G that are well suited for numerical evaluation can be found in the literature. In particular, complementary near-field and far-field asymptotic expansions and Taylor series

are given in Noblesse [7], Martin [8] and Telste & Noblesse [9]. Several practical approximate methods for computing G and ∇G have also been given. These alternative methods include polynomial approximations within complementary contiguous flow regions, given in Newman [10, 11], Chen [12, 13], Wang [14] and Zhou et al. [15], and table interpolation associated with function and coordinate transformations, given in Ba et al. [16] and Ponizy et al. [17]. Other useful practical methods can be found in the literature, notably in Peter & Meylan [18], Yao et al. [19], D'elía et al. [20] and Shen et al. [21].

Accuracy and efficiency are essential requirements of methods for numerically evaluating G and ∇G , and these important aspects are considered in the practical approximate methods listed in the foregoing. Indeed, the alternative methods proposed in these studies provide accurate and efficient methods for computing G and ∇G .

Numerical errors associated with potential-flow panel methods stem from several well-known sources, including:

- (i) discretization of the wetted hull surface of an offshore structure or a ship; i.e. the number and the type (flat or curved) of panels,
- (ii) approximation of the variations (piecewise constant, linear, quadratic, or higher-order) of the densities of the singularity (source, dipole) distributions over a surface panel,
- (iii) numerical integration of the Green function and its gradient over a (flat or curved) panel, and
- (iv) numerical approximation of the Green function and its gradient.

Moreover, the Green function G (as well as its gradient ∇G) is given by the sum of the fundamental free-space singularity, a wave component W and a non-oscillatory local flow component L , as was already noted. Thus, numerical errors that stem from the approximations of the local flow components in the representations of G and ∇G are only one part among several sources

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of errors associated with panel methods. While ideal approximations to G and ∇G should be highly accurate and efficient as well as very simple, this ideal goal is hard to reach in practice because accuracy, efficiency and simplicity are competing requirements.

The level of accuracy that is actually required for useful practical approximations to G and ∇G therefore is a fairly complicated issue. This issue is partly considered in Wu et al. [22] for the similar theory of steady ship waves (linear potential flow around a ship hull that advances at a constant speed in calm water). Specifically, the errors due to a simple analytical approximation to the local flow component L in the Green function for steady ship waves are considered in that study. This approximate local flow component L , given in Noblesse et al. [23], is very simple and highly efficient, but not particularly accurate. Yet, this simple relatively crude approximation to the Green function for steady ship waves is found in Wu et al. [22] to yield predictions of sinkage, trim and drag that do not differ appreciably from the predictions obtained if the Green function is computed with high accuracy. This finding suggests that highly accurate approximations to the local flow components in the Green function G and its gradient ∇G for the theory of wave diffraction radiation similarly may not be necessary for many practical purposes.

Simple approximations to the local flow components in the representations of G and ∇G for wave diffraction radiation are given here. These approximations are based on a pragmatic hybrid approach that combines numerical approximations with near-field and far-field analytical expansions, in a manner similar to that used in Noblesse et al. [23] for the Green function of the theory of ship waves. The approximations obtained here are valid within the entire flow region, i.e. are global approximations, unlike the approximations for complementary contiguous regions given in the literature. The approximations to the local flow components given here only involve elementary continuous functions (algebraic, exponential, logarithmic) of real arguments, and provide an efficient and particularly simple method for numerically evaluating the Green function G , and its gradient ∇G , for diffraction radiation of time-harmonic waves in deep water. The global approximations to the local flow components in G and ∇G given here are similar to, but considerably more accurate than, the approximations given in Wu et al. [24].

2. Basic integral representations

A Cartesian system of coordinates $\mathbf{X} \equiv (X, Y, Z)$ is used. The Z axis is vertical and points upward, and the undisturbed free surface is taken as the plane $Z = 0$. Diffraction radiation of time harmonic waves with radian frequency ω and wavelength $\lambda = 2\pi g/\omega^2$, where g denotes the gravitational acceleration, is considered. Nondimensional coordinates

$$\mathbf{x} \equiv (x, y, z) \equiv (X, Y, Z)\omega^2/g \quad (1)$$

are defined.

The Green function $G(\mathbf{x}, \tilde{\mathbf{x}})$ corresponds to the spatial component of a nondimensional velocity potential

$$\text{Re} [G(\mathbf{x}, \tilde{\mathbf{x}}) e^{-i\omega T}] \quad (2)$$

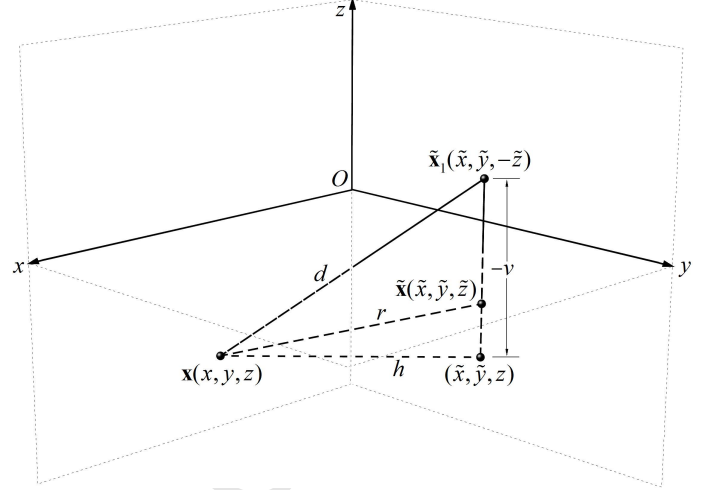


Figure 1: Definition sketch.

where T denotes time. Expression (2) represents the potential of the flow created at the point $\mathbf{x} \equiv (x, y, z \leq 0)$ by a pulsating source located at the point $\tilde{\mathbf{x}} \equiv (\tilde{x}, \tilde{y}, \tilde{z} < 0)$, or by a flux through the free surface at the point $\tilde{\mathbf{x}} \equiv (\tilde{x}, \tilde{y}, \tilde{z} = 0)$. Specifically, Noblesse [7, 25] show that the Green function satisfies the equations

$$\left. \begin{aligned} \nabla^2 G &= \delta(x - \tilde{x}) \delta(y - \tilde{y}) \delta(z - \tilde{z}) \text{ in } z < 0 \\ G_z - G &= 0 \text{ on } z = 0 \end{aligned} \right\} \text{ if } \tilde{z} < 0 \quad (3a)$$

$$\left. \begin{aligned} \nabla^2 G &= 0 \text{ in } z < 0 \\ G_z - G &= -\delta(x - \tilde{x}) \delta(y - \tilde{y}) \text{ on } z = 0 \end{aligned} \right\} \text{ if } \tilde{z} = 0 \quad (3b)$$

where $\delta(\cdot)$ stands for the usual Dirac delta function, and G_z is the z -derivative of the Green function G .

The nondimensional distances between the flow-field point $\mathbf{x}$ and the source point $\tilde{\mathbf{x}}$ or its mirror image $\tilde{\mathbf{x}}_1 \equiv (\tilde{x}, \tilde{y}, -\tilde{z})$ with respect to the undisturbed free-surface plane $z = 0$ are denoted as r and d , as is shown in Fig.1, and are given by

$$r \equiv \sqrt{(x - \tilde{x})^2 + (y - \tilde{y})^2 + (z - \tilde{z})^2} \quad (4a)$$

$$d \equiv \sqrt{(x - \tilde{x})^2 + (y - \tilde{y})^2 + (z + \tilde{z})^2} \quad (4b)$$

The horizontal and vertical components of the distance d between the points $\mathbf{x}$ and $\tilde{\mathbf{x}}_1$ are given by

$$0 \leq h \equiv \sqrt{(x - \tilde{x})^2 + (y - \tilde{y})^2} \text{ and } v \equiv z + \tilde{z} \leq 0 \quad (5)$$

The Green function G is expressed as

$$4\pi G = -1/r + L + W \quad (6)$$

where $-1/r$ is the fundamental free-space Green function, and L and W represent a local flow component and a wave component that account for free-surface effects. The component L corresponds to a non-oscillatory local flow and the component W represents circular surface waves radiated by the pulsating singularity located at the source point $\tilde{\mathbf{x}}$. The basic decomposition (6) into a local flow and waves is non unique, as was already noted. Indeed, three alternative decompositions and related integral representations are given in Noblesse [7].

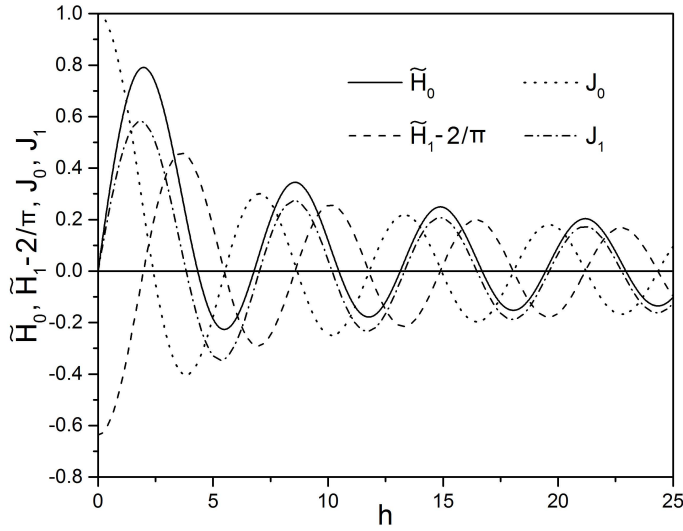


Figure 2: Functions $\tilde{H}_0(h)$, $\tilde{H}_1(h) - 2/\pi$, $J_0(h)$, $J_1(h)$ for $0 \leq h \leq 25$.

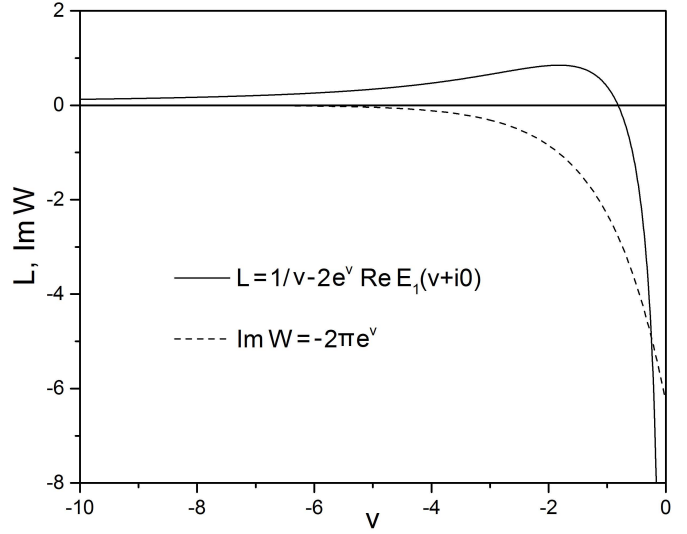


Figure 3: Local flow component L and imaginary part of the wave component $\text{Im } W$ for $h = 0$ and $-10 \leq v \leq 0$.

The so-called near-field integral representation in Noblesse [7] is considered here. The wave component W in this representation is given by

$$W(h, v) \equiv 2\pi [\tilde{H}_0(h) - i J_0(h)] e^v \quad (7)$$

where $\tilde{H}_0(\cdot)$ and $J_0(\cdot)$ denote the zeroth-order Struve function and the zeroth-order Bessel function of the first kind. The corresponding local flow component L is given by

$$L(h, v) \equiv -\frac{1}{d} - \frac{4}{\pi} \int_0^{\pi/2} \text{Re } e^M E_1(M) d\theta \text{ where } M \equiv v + i h \cos \theta \quad (8)$$

and $E_1(\cdot)$ is the usual complex exponential integral function.

The gradient $\nabla G \equiv (G_x, G_y, G_z)$ of the Green function G is expressed in Noblesse [7] as

$$4\pi G_z \equiv \frac{z - \tilde{z}}{r^3} + L_z + W \text{ where } L_z = \frac{v}{d^3} - \frac{1}{d} + L \quad (9a)$$

$$4\pi G_h \equiv \frac{h}{r^3} + L_h + W_h \text{ where } L_h = \frac{h}{d^3} + L_* \quad (9b)$$

$$4\pi G_x \equiv G_h \frac{x - \tilde{x}}{h} \text{ and } 4\pi G_y \equiv G_h \frac{y - \tilde{y}}{h} \quad (9c)$$

The wave component W_h in (9b) is given by

$$W_h(h, v) \equiv 2\pi [2/\pi - \tilde{H}_1(h) + i J_1(h)] e^v \quad (10)$$

where $\tilde{H}_1(\cdot)$ and $J_1(\cdot)$ denote the first-order Struve function and the first-order Bessel function of the first kind. The local flow component L_* in (9b) is given by

$$L_*(h, v) \equiv \frac{4}{\pi} \int_0^{\pi/2} \text{Im} [e^M E_1(M) - 1/M] \cos \theta d\theta \quad (11)$$

where M is defined by (8).

The exponential function e^v and the Bessel and Struve functions in expressions (7) and (10) for the wave components W and W_h are infinitely differentiable. Moreover, several practical and efficient alternative approximations for the Bessel and Struve functions are given in the literature; notably in Hitchcock [26], Abramowitz & Stegun [27], Luke [28], Newman [29]. Fig.2 depicts the Struve functions $\tilde{H}_0(h)$ and $\tilde{H}_1(h) - 2/\pi$ and the Bessel functions $J_0(h)$ and $J_1(h)$ for $0 \leq h \leq 25$.

3. Special cases

In the special case $h = 0$, expressions (7) and (10) for the wave components W and W_h become

$$W(h = 0, v) = -i 2\pi e^v \text{ and } W_h(h = 0, v) = 4 e^v \quad (12a)$$

Moreover, expression (8) yields $M = v$ and the integral representations (8) and (11) simplify as

$$L(h = 0, v) = 1/v - 2 e^v \text{Re } E_1(v + i0) \quad (12b)$$

$$L_*(h = 0, v) = -4 e^v \quad (12c)$$

Equations (9b), (12a) and (12c) then yield $G_h = 0$ for $h = 0$, in agreement with symmetry considerations. Fig.3 depicts the functions $L(h = 0, v)$ and $\text{Im } W(h = 0, v)$ for $-10 \leq v \leq 0$.

At the free-surface plane $v = 0$, expressions (4-7), (9) and (10) yield

$$4\pi \text{Re } G = -1/h + L(h, v = 0) + 2\pi \tilde{H}_0(h) = 4\pi \text{Re } G_z \quad (13a)$$

$$4\pi \text{Re } G_h = 2/h^2 + L_*(h, v = 0) + 2\pi [2/\pi - \tilde{H}_1(h)] \quad (13b)$$

The real parts $4\pi \text{Re } G$ and $4\pi \text{Re } G_z$ of the Green function G and its vertical derivative G_z at the free surface $v = 0$, and the related local flow and wave components

$$-1/h + L(h, v = 0) \text{ and } 2\pi \tilde{H}_0(h) \quad (14a)$$

are depicted in Fig.4 for $0 \leq h \leq 15$. Similarly, the real part $4\pi \text{Re } G_h$ of the horizontal derivative G_h of the Green function at the free surface $v = 0$, and the related local flow and wave components

$$2/h^2 + L_*(h, v = 0) \text{ and } 2\pi [2/\pi - \tilde{H}_1(h)] \quad (14b)$$

are depicted in Fig.5 for $0 \leq h \leq 15$.

Fig.4 and Fig.5 show that, at the free surface $v = 0$, the local flow components dominate the wave components W and W_h for $0 \leq h < 1$. However, the local flow components are significantly smaller than W and W_h for $15 \leq h$ or $4 \leq h$, respectively. The wave components W and W_h are infinitely differentiable and can readily be evaluated, as was already noted. Simple approximations to the local flow components L and L_* defined by the integral representations (8) and (11) are now considered.

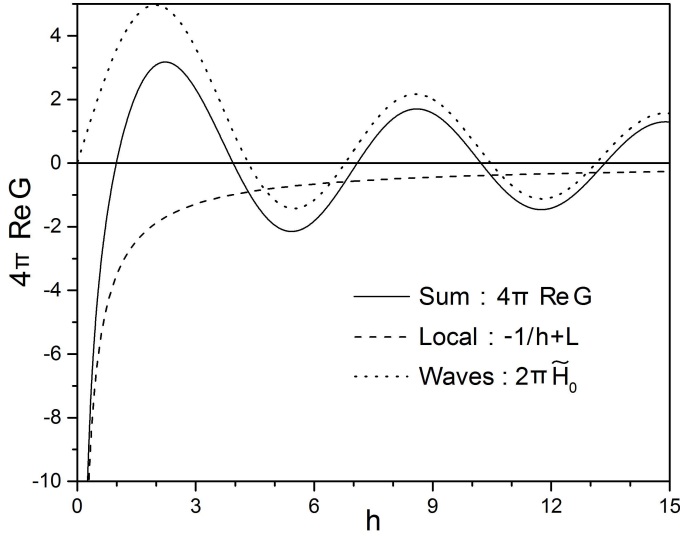


Figure 4: Real parts of the Green function and its vertical derivative, and related local flow and wave components, at the free surface $v = 0$ for $0 \leq h \leq 15$.

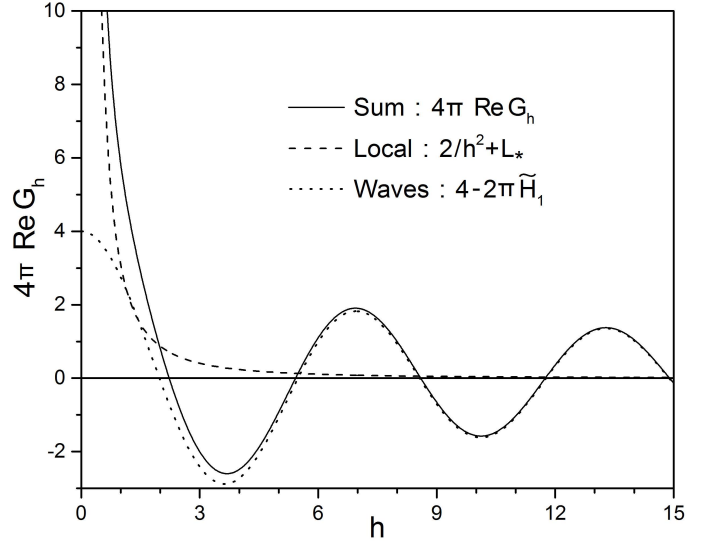


Figure 5: Real part of the horizontal derivative of the Green function and related local flow and wave components at the free surface $v = 0$ for $0 \leq h \leq 15$.

4. Near-field and far-field approximations

The variables $0 \leq \alpha \leq 1$ and $0 \leq \beta \leq 1$ defined as

$$\alpha \equiv -v/d \equiv \sqrt{1-\beta^2} \quad \text{and} \quad \beta \equiv h/d \equiv \sqrt{1-\alpha^2} \quad (15)$$

are used hereafter. The related variable $0 \leq \sigma \leq 1$ defined as

$$\sigma \equiv h/(d-v) \equiv \beta/(1+\alpha) \equiv \sqrt{(1-\alpha)/(1+\alpha)} \quad (16)$$

is also used in this section and the next.

The behaviors of the local flow component L defined by the integral representation (8) in the near-field and far-field limits $d \rightarrow 0$ and $d \rightarrow \infty$ are considered in Noblesse [7]. In particular, expressions (7.7), (7.8), (7.22) and (7.23) in Noblesse [7] yield the near-field approximation

$$L = -\frac{1}{d} + 2 \left(\log \frac{d-v}{2} + \gamma \right) + 2v \left(\log \frac{d-v}{2} + \gamma - 1 \right) + 2h(\sigma-2) + O(d^2 \log d) \quad \text{as } d \rightarrow 0 \quad (17)$$

where $\gamma = 0.577 \dots$ is Euler's constant. For $\alpha \neq 0$, expression (6.14) in Noblesse [7] yields the far-field approximation

$$L = \frac{1}{d} + \frac{2\alpha}{d^2} - \frac{2-6\alpha^2}{d^3} + O\left(\frac{1}{d^4}\right) \quad \text{as } d \rightarrow \infty \quad \text{if } \alpha \neq 0 \quad (18a)$$

For $\alpha = 0$, expressions (6.8) in Noblesse [7] and (12.1.30) in Abramowitz & Stegun [27] yield the far-field approximation

$$L = -\frac{3}{d} + \frac{2}{d^3} + O\left(\frac{1}{d^5}\right) \quad \text{as } d \rightarrow \infty \quad \text{if } \alpha = 0 \quad (18b)$$

Expression (9b) and the partial derivatives, with respect to h , of expressions (7.7), (7.8), (7.22) and (7.23) in Noblesse [7] yield the near-field approximation

$$L_* = \frac{2\sigma}{d} + 2(\sigma-2) - h \left(\log \frac{d-v}{2} + \gamma + \sigma\beta + 2\alpha - \frac{3}{2} \right) - 4v + O(d^2 \log d) \quad \text{as } d \rightarrow 0 \quad (19)$$

For $0 < \alpha < 1$, (9b) and (18a) similarly yield the far-field approximation

$$L_* = -\frac{2\beta}{d^2} - \frac{6\alpha\beta}{d^3} + O\left(\frac{1}{d^4}\right) \quad \text{as } d \rightarrow \infty \quad \text{if } 0 < \alpha < 1 \quad (20a)$$

In the special case $\alpha = 0$, equation (12.1.31) in Abramowitz & Stegun [27] and the derivative of (18b) yield the far-field approximation

$$L_* = \frac{2}{d^2} + O\left(\frac{1}{d^4}\right) \quad \text{as } d \rightarrow \infty \quad \text{if } \alpha = 0 \quad (20b)$$

In the special case $\alpha = 1$, one has $L_* \equiv -4e^v$ as was already noted in (12c).

5. Approach

The infinite flow region $0 \leq h < \infty$, $-\infty < v \leq 0$ is mapped onto the unit square

$$0 \leq \rho \equiv d/(1+d) \leq 1 \quad 0 \leq \beta \equiv h/d \leq 1 \quad (21a)$$

via the relations

$$d = \rho/(1-\rho) \quad h = \beta d \quad v = -\sqrt{1-\beta^2} d \quad (21b)$$

A pragmatic hybrid approach similar to that used in Noblesse et al. [23] for the Green function of steady ship waves is adopted here, as was already noted in the introduction. This approach combines numerical approximations with the near-field and far-field analytical expansions given in the previous section.

5.1. The function L

The local flow component L defined by (8) is expressed as

$$L = -\frac{1}{d} + \frac{2P}{1+d^3} + 2L' \quad (22a)$$

$$\text{where } P \equiv e^v \left(\log \frac{d-v}{2} + \gamma - 2d^2 \right) + d^2 - v \quad (22b)$$

$$\text{and } L' \equiv \frac{-P}{1+d^3} - \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \text{Re } e^M E_1(M) d\theta \quad (22c)$$

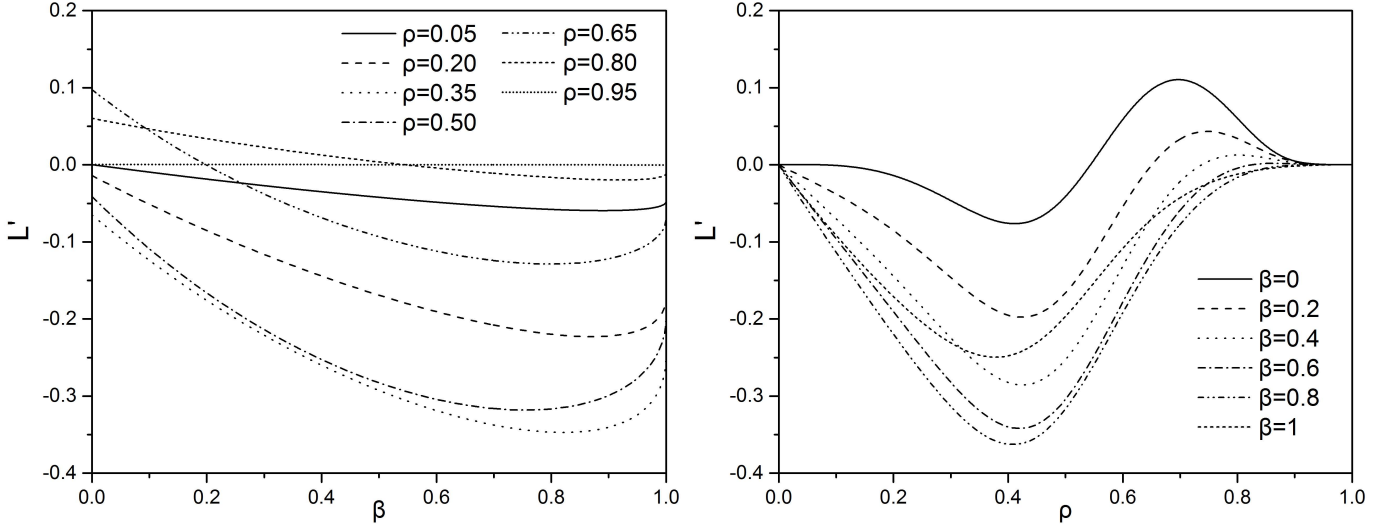


Figure 6: Variations of the function L' with respect to $0 \leq \beta \leq 1$ for $\rho = 0.05, 0.2, 0.35, 0.5, 0.65, 0.8, 0.95$ (left side) or with respect to $0 \leq \rho \leq 1$ for $\beta = 0, 0.2, 0.4, 0.6, 0.8, 1$ (right side).

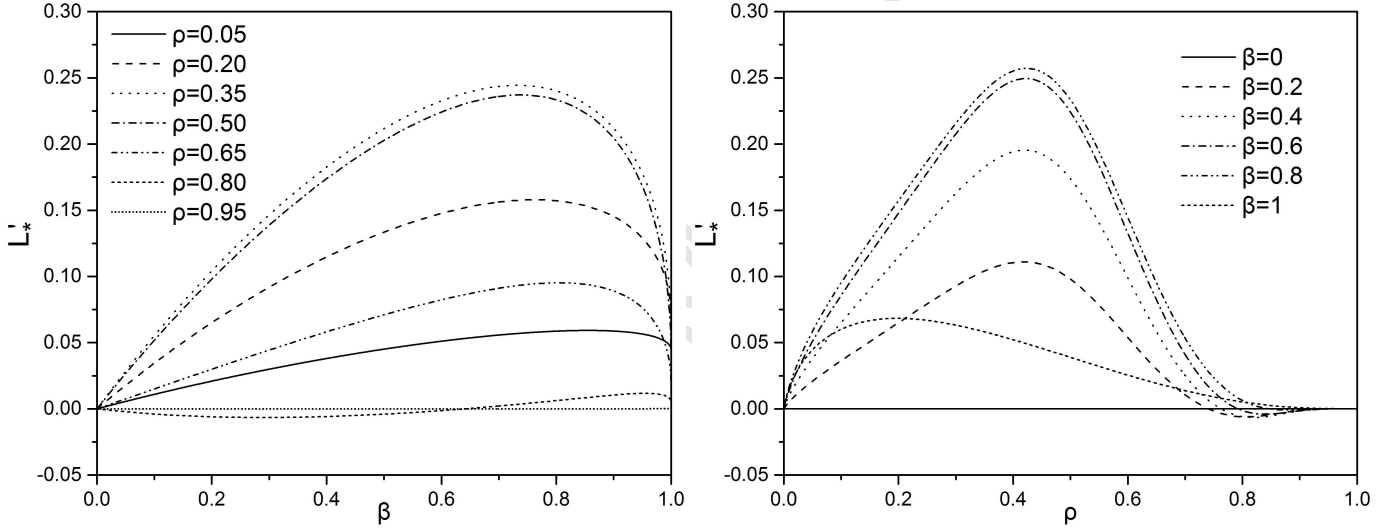


Figure 7: Variations of the function L'_* with respect to $0 \leq \beta \leq 1$ for $\rho = 0.05, 0.2, 0.35, 0.5, 0.65, 0.8, 0.95$ (left side) or with respect to $0 \leq \rho \leq 1$ for $\beta = 0, 0.2, 0.4, 0.6, 0.8, 1$ (right side).

Moreover, γ in (22b) is Euler's constant. The decomposition (22a) expresses the local flow L as the sum of two dominant terms that are defined analytically and are related to the near-field and far-field expansions (17) and (18), and the correction $2L'$. The correction term $2L'$ defined by (22c) is numerically approximated further on.

Eqs (17), (22a) and (22b) yield the near-field approximation

$$L' \sim d(\sigma\beta - 2\beta) \text{ as } d \rightarrow 0 \quad (23)$$

Eqs (18), (22a) and (22b) yield the far-field approximations

$$L' \sim \frac{1}{d^3}(3\alpha^2 - 1) \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 0 \quad (24a)$$

$$L' \sim -\frac{1}{d^3} \left(\log \frac{d}{2} + \gamma - 1 \right) \text{ as } d \rightarrow \infty \text{ if } \alpha = 0 \quad (24b)$$

The near-field approximations (17) and (23) and the far-

field approximations (18) and (24) yield

$$L'/L = O(d^2) \text{ as } d \rightarrow 0 \quad (25a)$$

$$L'/L = O(1/d^2) \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 0 \quad (25b)$$

$$L'/L = O(\log d/d^2) \text{ as } d \rightarrow \infty \text{ if } \alpha = 0 \quad (25c)$$

These relations show that $L' \ll L$ in both the near field $d \rightarrow 0$ and the far field $d \rightarrow \infty$. Expressions (21) and the asymptotic approximations (23) and (24) show that the function L' is asymptotically similar to $\rho(1-\rho)^3$ as $\rho \rightarrow 0$ and as $\rho \rightarrow 1$.

The variations of the function L' given by (22c) with respect to the variables β and ρ defined by (21) are depicted in Fig.6. An approximation to the function L' of the form

$$L'(\rho, \beta) \approx \rho(1-\rho)^3 R \text{ where} \quad (26a)$$

$$R \equiv (1-\beta)A - \beta B - \frac{\alpha C}{1+6\alpha\rho(1-\rho)} + \beta(1-\beta)D \quad (26b)$$

is considered in Section 6. The terms $A(\rho)$, $B(\rho)$, $C(\rho)$ and $D(\rho)$ in (26b) are polynomials in ρ .

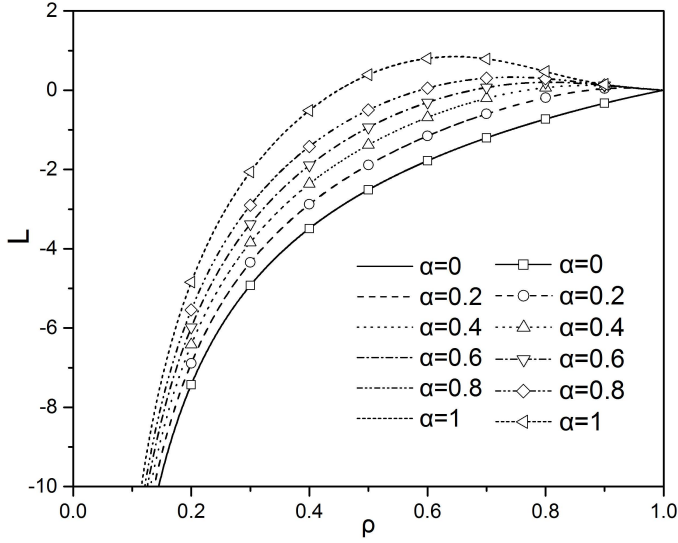


Figure 8: Local flow components L (lines without symbols) and L^a (lines with symbols) defined by the integral representation (8) or the related approximation (33) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

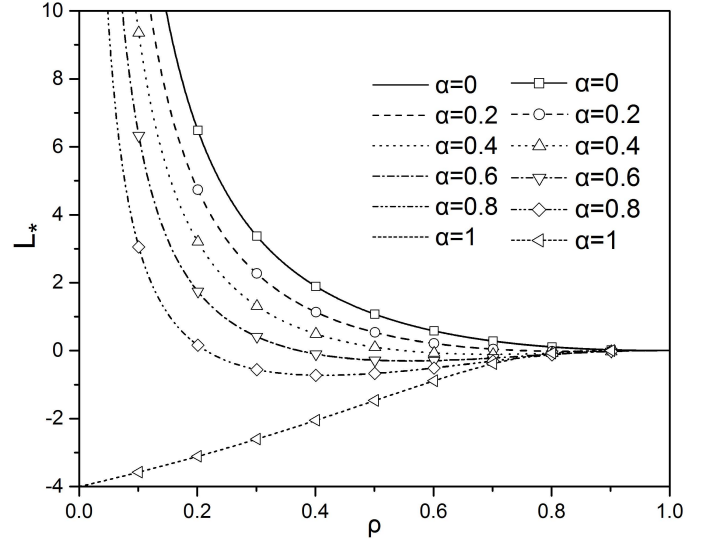


Figure 9: Local flow components L_* (lines without symbols) and L_*^a (lines with symbols) defined by the integral representation (11) or the related approximation (34) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

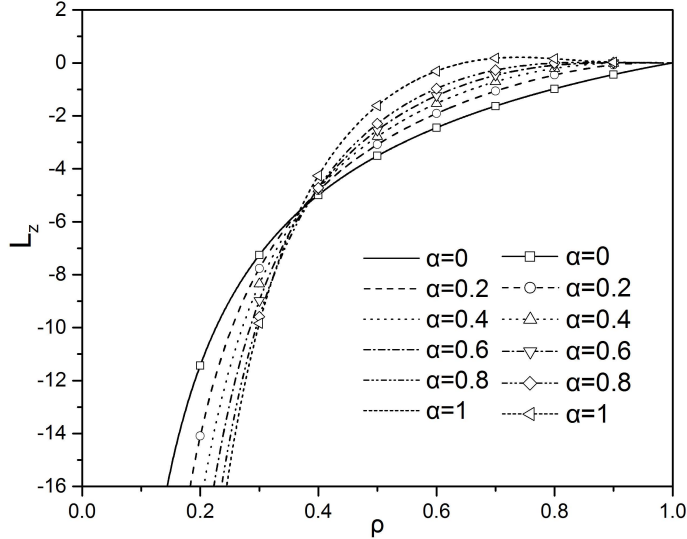


Figure 10: Local flow components L_z (lines without symbols) and L_z^a (lines with symbols) defined by the integral representation (8), (9a) or the related approximation (32b), (33) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

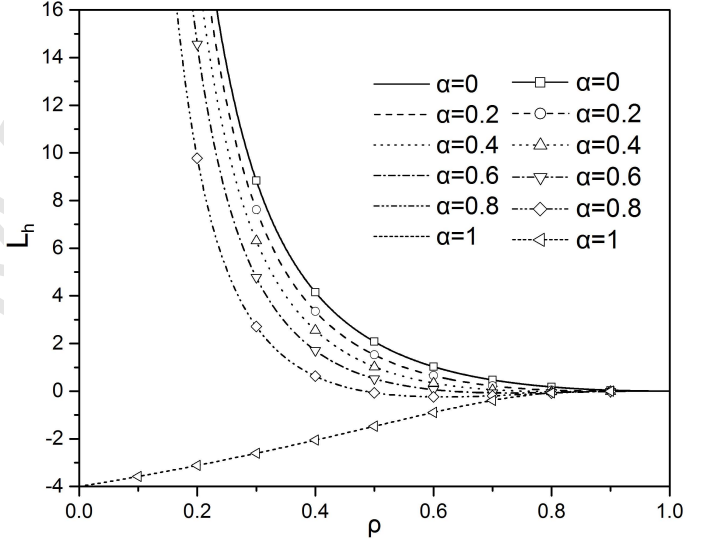


Figure 11: Local flow components L_h (lines without symbols) and L_h^a (lines with symbols) defined by the integral representation (9b), (11) or the related approximation (32b), (34) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

5.2. The function L_*

The local flow component L_* associated with the horizontal derivative G_h of the Green function G in (9b) is now considered. This flow component, defined by (11), is expressed as

$$L_* = \frac{2P_*}{1+d^3} - 4Q_* + 2L'_* \quad (27a)$$

$$\text{where } \begin{cases} P_* \equiv \frac{\beta+h}{d-v} - 2\beta + 2e^v d - h \\ Q_* \equiv e^{-d}(1-\beta) \left(1 + \frac{d}{1+d^3}\right) \end{cases} \quad (27b)$$

$$\text{and } L'_* \equiv \frac{-P_*}{1+d^3} + 2Q_* + \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \text{Im} \left[e^M E_1(M) - \frac{1}{M} \right] \cos \theta d\theta \quad (27c)$$

The decomposition (27a) expresses the local flow L_* as the sum of two dominant terms and the correction $2L'_*$ defined by (27c). This correction term is numerically approximated further on.

Eqs (19), (27a) and (27b) yield the near-field approximation

$$L'_* \sim -d \left[\frac{\beta}{2} \left(\log \frac{d-v}{2} + \gamma + \sigma\beta + 2\alpha - \frac{7}{2} \right) - 2\alpha + 2 \right] \text{ as } d \rightarrow 0 \quad (28)$$

Eqs (20), (27a) and (27b) yield the far-field approximation

$$L'_* \sim \frac{1}{d^3} (2\beta - \sigma - 3\alpha\beta) \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 1 \quad (29)$$

Moreover, expressions (12c), (27a) and (27b) yield $L'_* \equiv 0$ along the vertical axis $\alpha = 1$.

The near-field approximations (19), (28) and the far-field approximations (20) and (29) yield

$$L'_*/L_* = O(d^2 \log d) \text{ as } d \rightarrow 0 \quad (30a)$$

$$L'_*/L_* = O(1/d) \text{ as } d \rightarrow \infty \quad (30b)$$

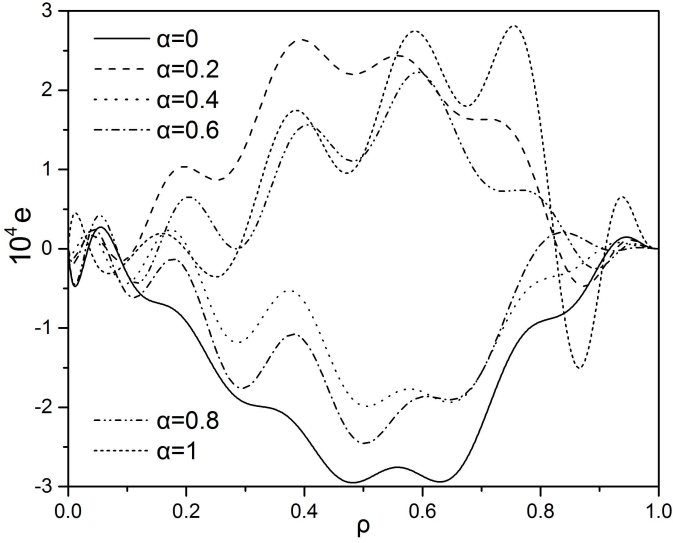


Figure 12: Absolute error $4\pi e \equiv L - L^a$ between the local flow components L and L^a defined by the integral representation (8) or the related approximation (33) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

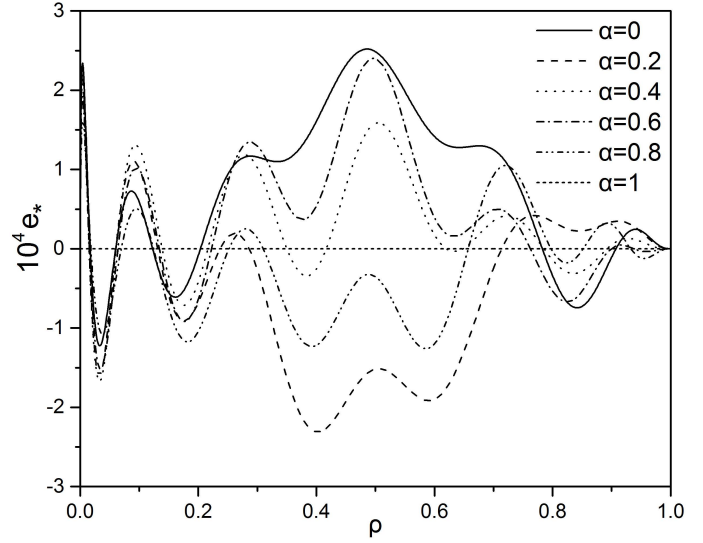


Figure 13: Absolute error $4\pi e^* \equiv L^* - L^{*a}$ between the local flow components L^* and L^{*a} defined by the integral representation (11) or the related approximation (34) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

where the identity $L'_* \equiv 0$ if $\alpha = 1$ was used. The relations (30) show that one has $L'_* \ll L_*$ in both the near field $d \rightarrow 0$ and the far field $d \rightarrow \infty$. Expressions (21) and the asymptotic approximations (28) and (29) show that the function L'_* is asymptotically similar to $\rho(1 - \rho)^3$ as $\rho \rightarrow 0$ and as $\rho \rightarrow 1$.

The variations of the function L'_* given by (27c) with respect to the variables β and ρ defined by (21) are depicted in Fig.7. The right side of Fig.7 shows that $L'_* \equiv 0$ along the vertical axis $\beta = 0$ as was already noted. An approximation to the function L'_* of the form

$$L'_*(\rho, \beta) \approx \rho(1 - \rho)^3 R_* \quad \text{where} \quad (31a)$$

$$R_* \equiv \beta A_* - (1 - \alpha) B_* + \beta(1 - \beta)\rho(1 - 2\rho) C_* \quad (31b)$$

is considered in the next section. The terms $A_*(\rho)$, $B_*(\rho)$ and $C_*(\rho)$ in (31b) are polynomials in ρ .

6. Practical approximations

The local flow components L and L_* defined by the integral representations (8) and (11) and the related local flow components L_z and L_h are approximated as

$$L \approx L^a \quad \text{and} \quad L_* \approx L_*^a \quad (32a)$$

$$L_z \approx L_z^a \equiv \frac{v}{d^3} - \frac{1}{d} + L^a \quad \text{and} \quad L_h \approx L_h^a \equiv \frac{h}{d^3} + L_*^a \quad (32b)$$

Hereafter, L^a , L_*^a , L_z^a and L_h^a denote approximations to the local flow components L , L_* , L_z and L_h , respectively.

Expressions (22a), (22b) and (26) define the approximate local flow component L^a as

$$L^a \equiv -\frac{1}{d} + \frac{2P}{1 + d^3} + 2\rho(1 - \rho)^3 R \quad (33a)$$

where P and R are defined by (22b) and (26b) as

$$P \equiv e^v \left(\log \frac{d - v}{2} + \gamma - 2d^2 \right) + d^2 - v \quad (33b)$$

$$R \equiv (1 - \beta)A - \beta B - \frac{\alpha C}{1 + 6\alpha\rho(1 - \rho)} + \beta(1 - \beta)D \quad (33c)$$

Here, $\gamma = 0.577 \dots$ is Euler's constant, and α, β, ρ are defined by (15) and (21a). Moreover, the polynomials $A(\rho)$, $B(\rho)$, $C(\rho)$ and $D(\rho)$ in (33c) are defined as

$$A \equiv 1.21 - 13.328\rho + 215.896\rho^2 - 1763.96\rho^3 + 8418.94\rho^4 - 24314.21\rho^5 + 42002.57\rho^6 - 41592.9\rho^7 + 21859\rho^8 - 4838.6\rho^9 \quad (33d)$$

$$B \equiv 0.938 + 5.373\rho - 67.92\rho^2 + 796.534\rho^3 - 4780.77\rho^4 + 17137.74\rho^5 - 36618.81\rho^6 + 44894.06\rho^7 - 29030.24\rho^8 + 7671.22\rho^9 \quad (33e)$$

$$C \equiv 1.268 - 9.747\rho + 209.653\rho^2 - 1397.89\rho^3 + 5155.67\rho^4 - 9844.35\rho^5 + 9136.4\rho^6 - 3272.62\rho^7 \quad (33f)$$

$$D \equiv 0.632 - 40.97\rho + 667.16\rho^2 - 6072.07\rho^3 + 31127.39\rho^4 - 96293.05\rho^5 + 181856.75\rho^6 - 205690.43\rho^7 + 128170.2\rho^8 - 33744.6\rho^9 \quad (33g)$$

Expressions (27a), (27b) and (31) define the approximate local flow component L_*^a as

$$L_*^a \equiv \frac{2P_*}{1 + d^3} - 4Q_* + 2\rho(1 - \rho)^3 R_* \quad (34a)$$

where P_* , Q_* and R_* are defined by (27b) and (31b) as

$$P_* \equiv \frac{\beta + h}{d - v} - 2\beta + 2e^v d - h \quad (34b)$$

$$Q_* \equiv e^{-d}(1 - \beta) \left(1 + \frac{d}{1 + d^3} \right) \quad (34c)$$

$$R_* \equiv \beta A_* - (1 - \alpha) B_* + \beta(1 - \beta)\rho(1 - 2\rho) C_* \quad (34d)$$

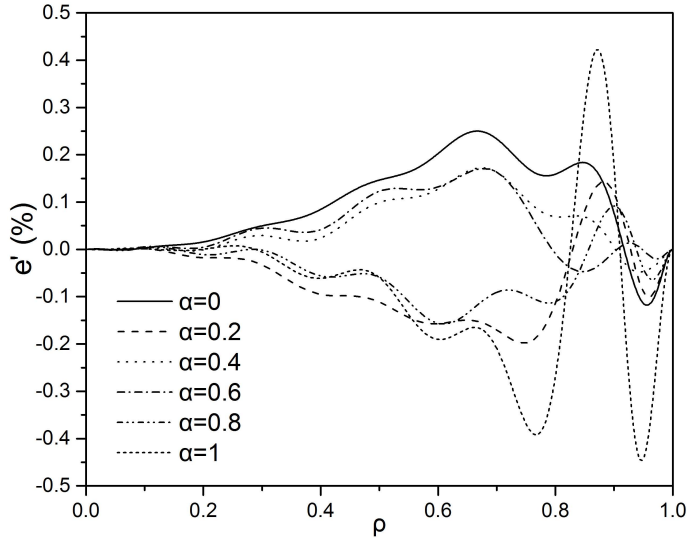


Figure 14: Approximate relative error $e' \equiv (L - L^a)/L^{v=0}$ between the local flow components L and L^a defined by the integral representation (8) or the approximation (33) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

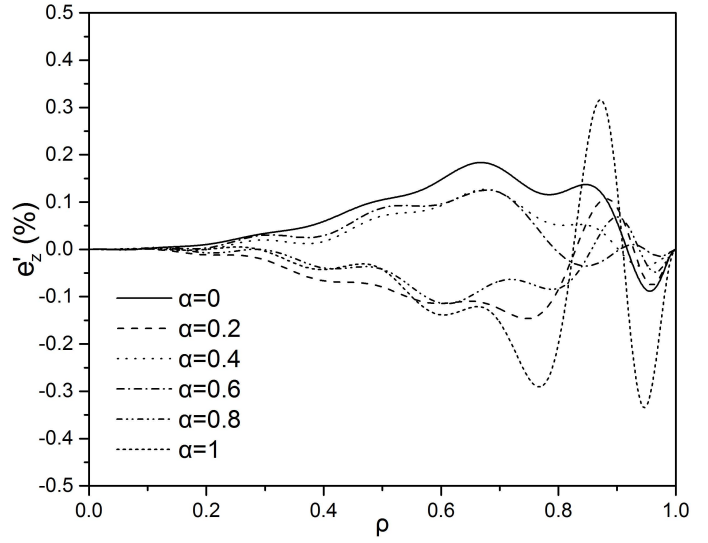


Figure 15: Approximate relative error $e'_z \equiv (L_z - L_z^a)/L_z^{v=0}$ between the local flow components L_z and L_z^a defined by the integral representation (8), (9a) or the approximation (32b), (33) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

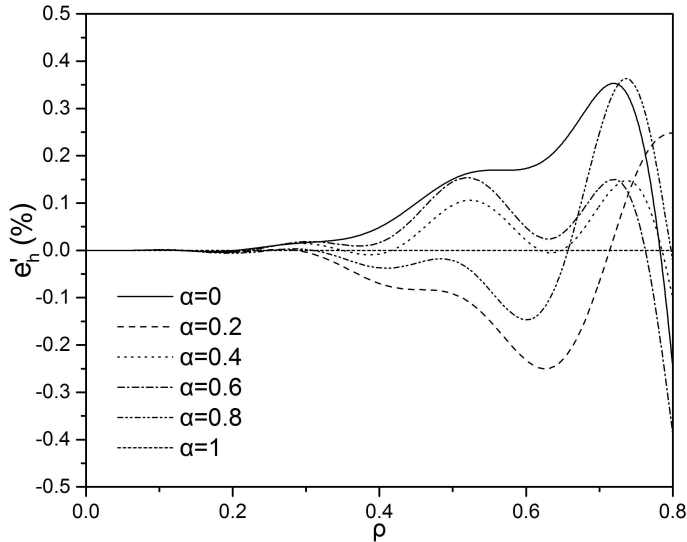


Figure 16: Approximate relative error $e'_h \equiv (L_h - L_h^a)/L_h^{v=0}$ between the local flow components L_h and L_h^a defined by the integral representation (9b), (11) or the approximation (32b), (34) for $0 \leq \rho \leq 0.8$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

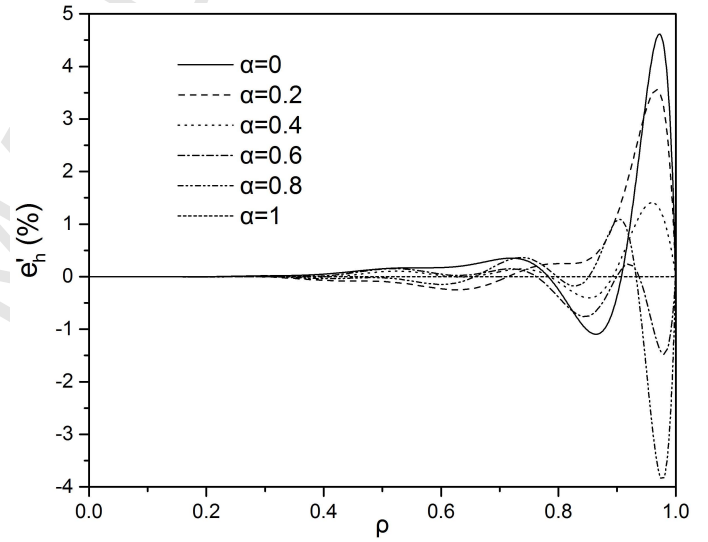


Figure 17: Approximate relative error $e'_h \equiv (L_h - L_h^a)/L_h^{v=0}$ between the local flow components L_h and L_h^a defined by the integral representation (9b), (11) or the approximation (32b), (34) for $0 \leq \rho \leq 1$ and $\alpha = 0, 0.2, 0.4, 0.6, 0.8, 1$.

The polynomials $A_*(\rho)$, $B_*(\rho)$ and $C_*(\rho)$ in (34d) are defined as

$$A_* \equiv 2.948 - 24.53\rho + 249.69\rho^2 - 754.85\rho^3 - 1187.71\rho^4 \\ + 16370.75\rho^5 - 48811.41\rho^6 + 68220.87\rho^7 - 46688\rho^8 \\ + 12622.25\rho^9 \quad (34e)$$

$$B_* \equiv 1.11 + 2.894\rho - 76.765\rho^2 + 1565.35\rho^3 - 11336.19\rho^4 \\ + 44270.15\rho^5 - 97014.11\rho^6 + 118879.26\rho^7 \\ - 76209.82\rho^8 + 19923.28\rho^9 \quad (34f)$$

$$C_* \equiv 14.19 - 148.24\rho + 847.8\rho^2 - 2318.58\rho^3 + 3168.35\rho^4 \\ - 1590.27\rho^5 \quad (34g)$$

The approximations L^a and L_*^a given by (33) and (34) hold within the entire flow region $0 \leq d$, and only involve elementary continuous functions (algebraic, exponential, logarithmic) of real arguments.

Fig.8 depicts the local flow components L and L^a defined

by the integral representation (8) or the related approximation (33) for $0 \leq \rho \leq 1$ and six values of $0 \leq \alpha \leq 1$. Fig.9 similarly depicts the local flow components L_* and L_*^a given by the integral representation (11) or the related approximation (34) for $0 \leq \rho \leq 1$ and six values of $0 \leq \alpha \leq 1$.

Fig.10 depicts the local flow components L_z and L_z^a defined by the integral representation (8), (9a) or the related approximation (32b), (33) for $0 \leq \rho \leq 1$ and six values of $0 \leq \alpha \leq 1$. Fig.11 similarly depicts the local flow components L_h and L_h^a given by the integral representation (9b), (11) or the related approximation (32b), (34) for $0 \leq \rho \leq 1$ and six values of $0 \leq \alpha \leq 1$.

The functions L and L^a in Fig.8, the functions L_* and L_*^a in Fig.9, the functions L_z and L_z^a in Fig.10, and the functions L_h and L_h^a in Fig.11 cannot be distinguished.

7. Errors in the Green function and its gradient

The errors associated with the approximations L^a and L_* given by (33) and (34) are now considered.

7.1. Definition of absolute and relative errors

The absolute errors e and e_* between the local flow components L and L_* and the corresponding approximations L^a and L_* in expressions (6) and (9) for the Green function and its gradient are defined as

$$4\pi e \equiv L - L^a \equiv L_z - L_z^a \quad (35a)$$

$$4\pi e_* \equiv L_* - L_*^a \equiv L_h - L_h^a \quad (35b)$$

As was already noted, one has $L_*^a \equiv L_*$ along the vertical axis $\alpha = 1$, and (35b) then yields

$$e_* \equiv 0 \text{ if } \alpha = 1 \quad (36)$$

The relative errors associated with the approximate local flow components L^a , L_z^a and L_h^a are defined as

$$e^r \equiv (L - L^a)/L \equiv 4\pi e/L \quad (37a)$$

$$e_z^r \equiv (L_z - L_z^a)/L_z \equiv (L - L^a)/L_z \equiv 4\pi e/L_z \quad (37b)$$

$$e_h^r \equiv (L_h - L_h^a)/L_h \equiv (L_* - L_*^a)/L_h \equiv 4\pi e_*/L_h \quad (37c)$$

Figs 8, 10 and 11 show that the functions $L(h, v)$, $L_z(h, v)$ and $L_h(h, v)$ vanish at a point $\rho < 1$ if $v < 0$, but do not vanish for $\rho < 1$ if $v = 0$. The relative errors e^r , e_z^r and e_h^r defined by (37) are then approximated here via the modified relative errors e' , e'_z and e'_h defined as

$$e' \equiv \frac{L - L^a}{L^{v=0}}, \quad e'_z \equiv \frac{L_z - L_z^a}{L_z^{v=0}} \equiv e' \frac{L^{v=0}}{L_z^{v=0}}, \quad e'_h \equiv \frac{L_h - L_h^a}{L_h^{v=0}} \quad (38)$$

where $L^{v=0} \equiv L(h, v = 0)$, and $L_z^{v=0}$ and $L_h^{v=0}$ similarly denote the functions $L_z(h, v = 0)$ and $L_h(h, v = 0)$. The functions e' , e'_z and e'_h defined by (38) provide meaningful approximations to the relative errors associated with the approximations L^a , L_z^a and L_h^a within the entire flow region $0 \leq d$.

Expression (9a) yields $L_z^{v=0} = L^{v=0} - 1/h$. This expression and the asymptotic approximations (17) and (18b) then yield

$$\frac{L^{v=0}}{L_z^{v=0}} \approx \frac{1}{2} \text{ as } h \rightarrow 0 \text{ and } \frac{L^{v=0}}{L_z^{v=0}} \approx \frac{3}{4} - \frac{1}{8h^2} \text{ as } h \rightarrow \infty \quad (39)$$

These relations show that the (approximate) relative errors e'_z and e' in (38) are closely related, and that one has $e'_z \approx 3e'/4$ over most of the flow region, as is illustrated further on in Fig.14 and Fig.15.

7.2. Asymptotic analysis of absolute and relative errors

Equations (17), (18) and (33) show that the absolute error e defined by (35a) behaves as

$$e = O(d) \text{ as } d \rightarrow 0 \quad (40a)$$

$$e = O(1/d^3) \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 0 \quad (40b)$$

$$e = O(\log d/d^3) \text{ as } d \rightarrow \infty \text{ if } \alpha = 0 \quad (40c)$$

Equations (19), (20), (34), (35b) and (36) similarly yield

$$e_* = O(d \log d) \text{ as } d \rightarrow 0 \quad (41a)$$

$$e_* = O(1/d^3) \text{ as } d \rightarrow \infty \quad (41b)$$

The relative errors (37) are now considered. Equations (9a), (17) and (18) yield

$$L_z \sim -\alpha/d^2 \text{ as } d \rightarrow 0 \text{ if } \alpha \neq 0 \quad (42a)$$

$$L_z \sim -2/d \text{ as } d \rightarrow 0 \text{ if } \alpha = 0 \quad (42b)$$

$$L_z \sim \alpha/d^2 \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 0 \quad (42c)$$

$$L_z \sim -4/d \text{ as } d \rightarrow \infty \text{ if } \alpha = 0 \quad (42d)$$

Similarly, equations (9b), (19) and (20) yield

$$L_h \sim \beta/d^2 \text{ as } d \rightarrow 0 \text{ if } \alpha \neq 1 \quad (43a)$$

$$L_h \rightarrow -4 \text{ as } d \rightarrow 0 \text{ if } \alpha = 1 \quad (43b)$$

$$L_h \sim -\beta/d^2 \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 0 \text{ or } 1 \quad (43c)$$

$$L_h \sim 3/d^2 \text{ as } d \rightarrow \infty \text{ if } \alpha = 0 \quad (43d)$$

Moreover, equations (9b) and (12c) yield $L_h = -4e^v$ if $\alpha = 1$.

The asymptotic approximations (17), (18) and (40) show that the relative error e^r defined by (37a) behaves as

$$e/L = O(d^2) \text{ as } d \rightarrow 0 \quad (44a)$$

$$e/L = O(1/d^2) \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 0 \quad (44b)$$

$$e/L = O(\log d/d^2) \text{ as } d \rightarrow \infty \text{ if } \alpha = 0 \quad (44c)$$

One then has $e \ll L$ in both the near field and the far field. Expressions (40) and (42) similarly show that the relative error e_z^r defined by (37b) behaves as

$$e/L_z = O(d^3) \text{ as } d \rightarrow 0 \text{ if } \alpha \neq 0 \quad (45a)$$

$$e/L_z = O(d^2) \text{ as } d \rightarrow 0 \text{ if } \alpha = 0 \quad (45b)$$

$$e/L_z = O(1/d) \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 0 \quad (45c)$$

$$e/L_z = O(\log d/d^2) \text{ as } d \rightarrow \infty \text{ if } \alpha = 0 \quad (45d)$$

One then has $e \ll L_z$ in both the near field and the far field. Expressions (41) and (43) likewise show that the relative error e_h^r defined by (37c) behaves as

$$e_*/L_h = O(d^3 \log d) \text{ as } d \rightarrow 0 \text{ if } \alpha \neq 1 \quad (46a)$$

$$e_*/L_h = O(1/d) \text{ as } d \rightarrow \infty \text{ if } \alpha \neq 1 \quad (46b)$$

Along the vertical axis $\alpha = 1$, one has $e_* \equiv 0$ in accordance with (36). One then has $e_* \ll L_h$ in both the near field and the far field.

7.3. Numerical analysis of absolute and relative errors

Computations within the entire region $0 \leq d$ show that the absolute errors e and e_* defined by (35) vary within the ranges

$$-3 \times 10^{-4} < e < 3 \times 10^{-4}, \quad -2.4 \times 10^{-4} < e_* < 2.6 \times 10^{-4} \quad (47)$$

Fig.12 and Fig.13 depict the errors e and e_* for $0 \leq \rho \leq 1$ and six values of $0 \leq \alpha \leq 1$. Fig.12 and Fig.13 show that the errors e and e_* vanish both as $\rho \rightarrow 0$ and as $\rho \rightarrow 1$, i.e. in the near field $d \rightarrow 0$ and the far field $d \rightarrow \infty$, in accordance with (40) and (41) and the fact that the global approximations

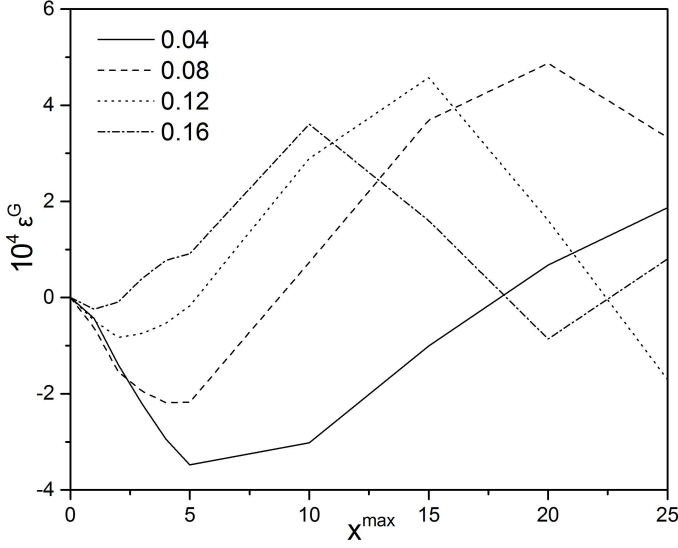


Figure 18: Relative error ε^G defined by (51) for $1 \leq x^{\max} \leq 25$ and $-z^{\max}/x^{\max} = 0.04, 0.08, 0.12, 0.16$.

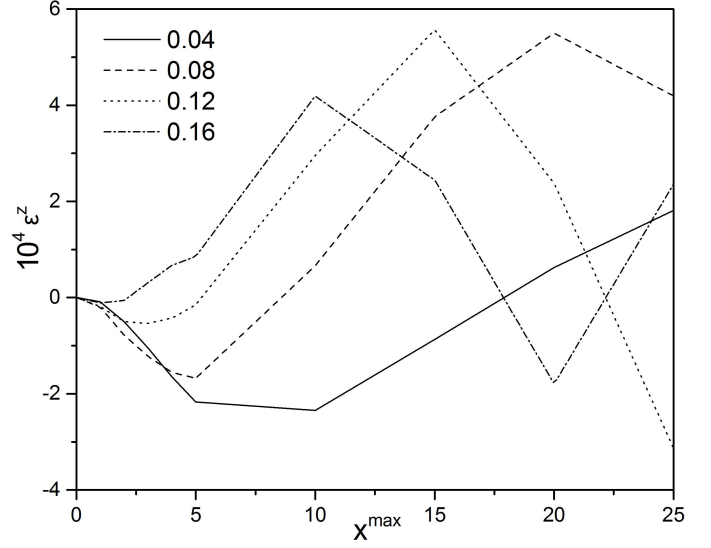


Figure 19: Relative error ε^z defined by (51) for $1 \leq x^{\max} \leq 25$ and $-z^{\max}/x^{\max} = 0.04, 0.08, 0.12, 0.16$.

L^a and L_*^a given by (33) and (34) are asymptotically correct in the limits $d \rightarrow 0$ and $d \rightarrow \infty$. Specifically, Fig.12 shows that the absolute error e associated with the approximations L^a and L_z^a is relatively small in the near field $0 \leq \rho \leq 0.15$ and in the far field $0.81 \leq \rho \leq 1$, Fig.13 shows that the absolute error e_* associated with the approximation L_h^a is relatively small in the far field $0.75 \leq \rho \leq 1$.

The errors e and e_* for the six values of α considered in Fig.12 and Fig.13 are of the same order of magnitude, and oscillate between positive and negative values that are distributed more or less evenly. Fig.13 shows that $e_* \equiv 0$ along the vertical axis $\alpha = 1$, in accordance with (36).

Fig.14 and Fig.15 depict the approximate relative errors e' and e'_z for $0 \leq \rho \leq 1$ and six values of $0 \leq \alpha \leq 1$. Computations within the entire region $0 \leq d$ show that e' and e'_z vary within the ranges

$$-0.45 \times 10^{-2} < e' < 0.43 \times 10^{-2} \quad (48a)$$

$$-0.34 \times 10^{-2} < e'_z < 0.32 \times 10^{-2} \quad (48b)$$

Fig.16 and Fig.17 similarly depict the approximate relative error e'_h for $0 \leq \rho \leq 0.8$ or $0 \leq \rho \leq 1$ and the same six values of $0 \leq \alpha \leq 1$. The errors e' , e'_z and e'_h in Fig.14-Fig.16 are relatively small and of the same order of magnitude. Fig.17 shows that the error e'_h is significantly larger in the far field $0.8 \leq \rho \leq 1$. Indeed, computations within the entire region $0 \leq d$ show that e'_h vary within the range

$$-4.33 \times 10^{-2} < e'_h < 4.73 \times 10^{-2} \quad (48c)$$

However, Fig.13 shows that the absolute error e_* that corresponds to the relative error e'_h is small in the range $0.8 \leq \rho \leq 1$ within which e'_h is large. Specifically, Fig.13 shows that the absolute error $|e_*|$ is smaller than 10^{-4} for $0.8 \leq \rho \leq 0.9$ and smaller than 0.5×10^{-4} for $0.9 \leq \rho \leq 1$.

Fig.14-Fig.17 show that the relative errors e' , e'_z and e'_h vanish in both the near field $\rho \rightarrow 0$ and the far field $\rho \rightarrow 1$. These errors are very small in the near field, where the functions L , L_z and L_h are large as is shown in Fig.8, Fig.10 and Fig.11. Moreover, Fig.16 and Fig.17 show that one has $e_* \equiv 0$ and therefore $e'_h \equiv 0$ along the vertical axis $\alpha = 1$.

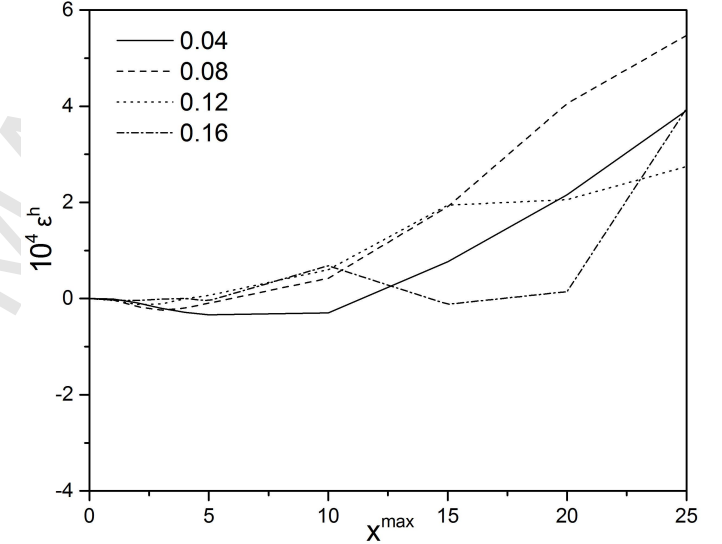


Figure 20: Relative error ε^h defined by (51) for $1 \leq x^{\max} \leq 25$ and $-z^{\max}/x^{\max} = 0.04, 0.08, 0.12, 0.16$.

7.4. Numerical analysis of global errors

The errors associated with the analytical approximations G^a , G_z^a and G_h^a to the Green function G and its derivatives G_z and G_h can be further analyzed by considering flow field points $\mathbf{x}_{ij} \equiv (x_i, 0, z_j)$ and source points $\tilde{\mathbf{x}}_{mn} \equiv (\tilde{x}_m, 0, \tilde{z}_n)$ where x_i , z_j , $\tilde{x}_m$ and $\tilde{z}_n$ are defined as

$$x_i \equiv (i-1)\delta^x \text{ with } 1 \leq i \leq N^x + 1 \quad (49a)$$

$$z_j \equiv -(j-1)\delta^z \text{ with } 1 \leq j \leq N^z + 1 \quad (49b)$$

$$\tilde{x}_m \equiv (m-1/2)\delta^x \text{ with } 1 \leq m \leq N^x \quad (49c)$$

$$\tilde{z}_n \equiv -(n-1/2)\delta^z \text{ with } 1 \leq n \leq N^z \quad (49d)$$

The flow field points $\mathbf{x}_{ij}$ are knots of a rectangular grid within the region $0 \leq x \leq x^{\max} \equiv N^x \delta^x$, $-z^{\max} \equiv N^z \delta^z \leq z \leq 0$. The grid size is chosen as $\delta^x = 0.01$ and $\delta^z = 0.01$ for $x^{\max} \leq 5$, or as $\delta^x = 0.1$ and $\delta^z = 0.1$ for $5 < x^{\max}$. The source points $\tilde{\mathbf{x}}_{mn}$ are the centroids of the panels defined by the rectangular grid associated with the flow field points $\mathbf{x}_{ij}$.

The integrated Green function $\bar{G}$ and the related functions $\bar{G}_z$ and $\bar{G}_h$ defined as

$$\bar{G} \equiv \sum_{i=1}^{N^x+1} \sum_{j=1}^{N^z+1} \sum_{m=1}^{N^x} \sum_{n=1}^{N^z} G(\mathbf{x}_{ij}, \tilde{\mathbf{x}}_{mn}) \quad (50a)$$

$$\bar{G}_z \equiv \sum_{i=1}^{N^x+1} \sum_{j=1}^{N^z+1} \sum_{m=1}^{N^x} \sum_{n=1}^{N^z} G_z(\mathbf{x}_{ij}, \tilde{\mathbf{x}}_{mn}) \quad (50b)$$

$$\bar{G}_h \equiv \sum_{i=1}^{N^x+1} \sum_{j=1}^{N^z+1} \sum_{m=1}^{N^x} \sum_{n=1}^{N^z} G_h(\mathbf{x}_{ij}, \tilde{\mathbf{x}}_{mn}) \quad (50c)$$

are considered together with the integrated Green function $\bar{G}^a$ and the related functions $\bar{G}_z^a$ and $\bar{G}_h^a$ defined by (50) where the Green function G and its derivatives G_z and G_h are replaced by their corresponding approximations G^a , G_z^a and G_h^a .

The relative errors

$$\varepsilon^G \equiv 1 - \bar{G}^a/\bar{G}, \quad \varepsilon^z \equiv 1 - \bar{G}_z^a/\bar{G}_z, \quad \varepsilon^h \equiv 1 - \bar{G}_h^a/\bar{G}_h \quad (51)$$

provide an estimate of the relative errors that may be expected in numerical computations of added-mass and wave-damping coefficients based on a panel method as a result of the approximations G^a , G_z^a and G_h^a to the Green function G and its derivatives. The relative errors ε^G , ε^z and ε^h defined by (51), where the wave components W and W_h in expressions (6), (9a) and (9b) are ignored because the components W and W_h can be evaluated with very high accuracy, are considered here to estimate the global integrated errors that may be expected as a result of the approximations to the local flow components in the Green function G and its gradient.

Fig.18, Fig.19 and Fig.20 depict the relative errors ε^G , ε^z and ε^h defined by (51) for flow regions defined by nine values of x^{max} within the range $1 \leq x^{max} \leq 25$ and four values of $-z^{max}/x^{max}$ taken as 0.04, 0.08, 0.12 and 0.16. These three figures show that the relative errors ε^G , ε^z and ε^h vary between positive and negative values within the ranges

$$-4 \times 10^{-4} < \varepsilon^G < 5 \times 10^{-4} \quad (52a)$$

$$-4 \times 10^{-4} < \varepsilon^z < 6 \times 10^{-4} \quad (52b)$$

$$-1 \times 10^{-4} < \varepsilon^h < 6 \times 10^{-4} \quad (52c)$$

These errors are relatively small, especially for small values of x^{max} , and are of the same order of magnitude for the four values of $-z^{max}/x^{max}$.

8. Conclusion

The Green function G in the classical theory of wave diffraction radiation by an offshore structure, or a ship at low forward speed, in deep water is expressed in the usual manner as the sum of the fundamental free-space Green function $-1/r$, a non-oscillatory local flow component L and a wave component W . The gradient of G is similarly expressed as the sum of three basic components. The wave components W and W_h in these basic decompositions of G and its gradient ∇G are expressed in terms of real functions of one variable, specifically the exponential function e^v , the Bessel functions $J_0(h)$ and $J_1(h)$ and the Struve functions $\tilde{H}_0(h)$ and $\tilde{H}_1(h)$. These functions are infinitely differentiable and can be readily evaluated; e.g. Hitchcock [26], Abramowitz & Stegun [27], Luke [28], Newman [29].

Expressions (6), (7), (9), (10) and (32)–(34) provide a practical basis for evaluating the Green function G and its gradient ∇G . The analytical approximations (33) and (34) to the local

flow components in the expressions for G and ∇G are global approximations valid within the entire flow region, unlike the approximations for complementary contiguous flow regions given in the literature. The global approximations obtained here only involve elementary continuous functions (algebraic, exponential, logarithmic) of real arguments. The analysis of the errors associated with the approximations to the local flow components given in the study shows that the approximations are sufficiently accurate for practical purposes. These global approximations provide a particularly simple and highly efficient way of numerically evaluating the Green function and its gradient for diffraction radiation of time-harmonic waves in deep water.

- [1] T.H. Havelock, The damping of the heaving and pitching motion of a ship, The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science 33 (224) (1942) 666-673.
- [2] T.H. Havelock, Waves due to a floating sphere making periodic heaving oscillations, Proceedings of the Royal Society of London A: Mathematical, Physical and Engineering Sciences 231 (1184) (1955) 1-7.
- [3] R.C. Thorne, Multipole expansions in the theory of surface waves, Mathematical Proceedings of the Cambridge Philosophical Society 49 (4) (1953) 707-716.
- [4] M.D. Haskind, On wave motion of a heavy fluid, Prikladnaya Matematika i Mekhanika 18 (1954) 15-26.
- [5] J.V. Wehausen, E.V. Laitone, Surface waves, Handbuch der Physik, Volume 9, Springer-Verlag, Berlin (1960) 446-778.
- [6] W.D. Kim, On the harmonic oscillations of a rigid body on a free surface, Journal of Fluid Mechanics 21 (3) (1965) 427-451.
- [7] F. Noblesse, The Green function in the theory of radiation and diffraction of regular water waves by a body, Journal of Engineering Mathematics 16 (2) (1982) 137-169.
- [8] D. Martin, Résolution numérique du problème linéarisé de la tenue à la mer, Association Technique Maritime et Aéronautique (1980).
- [9] J.G. Telste, F. Noblesse, Numerical evaluation of the Green function of water-wave radiation and diffraction, Journal of Ship Research 30 (2) (1986) 69-84.
- [10] J.N. Newman, An expansion of the oscillatory source potential, Applied Ocean Research 6 (2) (1984) 116-117.
- [11] J.N. Newman, Algorithms for the free-surface Green function, Journal of Engineering Mathematics 19 (1) (1985) 57-67.
- [12] X.B. Chen, Free surface Green function and its approximation by polynomial series, Bureau Veritas' Research Report No. 641 DTO/XC, Bureau Veritas, France, (1991).
- [13] X.B. Chen, Evaluation de la fonction de Green du problème de diffraction/radiation en profondeur d'eau finie-une nouvelle méthode rapide et précise, 4e Journées de l'Hydrodynamique, Nantes, France, (1993) 371-384.
- [14] R.S. Wang, The numerical approach of three dimensional free-surface Green function and its derivatives, Journal of Hydrodynamics (Ser.A) 7 (3) (1992) 277-286.
- [15] Q. Zhou, G. Zhang, L. Zhu, The fast calculation of free-surface wave Green function and its derivatives, Chinese Journal of Computational Physics 16 (2) (1999) 113-120.
- [16] M. Ba, B. Ponizy, F. Noblesse, Calculation of the Green function of water-wave diffraction and radiation, The Second International Offshore and Polar Engineering Conference, San Francisco, USA, (1992) 60-63.
- [17] B. Ponizy, F. Noblesse, M. Ba, M. Guilbaud, Numerical evaluation of free-surface Green functions, Journal of Ship Research 38 (3) (1994) 193-202.
- [18] M.A. Peter, M.H. Meylan, The eigenfunction expansion of the infinite depth free surface Green function in three dimensions, Wave Motion 40 (1) (2004) 1-11.
- [19] X.L. Yao, S.L. Sun, S.P. Wang, S.T. Yang, The research on the highly efficient calculation method of 3-D frequency-domain Green function, Journal of Marine Science and Application 8 (3) (2009) 196-203.
- [20] J. D'elía, L. Battaglia, M. Storti, A semi-analytical computation of the Kelvin kernel for potential flows with a free surface, Computational & Applied Mathematics 30 (2) (2011) 267-287.
- [21] Y. Shen, D. Yu, W. Duan, H. Ling, Ordinary differential equation algorithms for a frequency-domain water wave Green's function, Journal of Engineering Mathematics 100 (1) (2016) 53-66.
- [22] H. Wu, C. Zhang, C. Ma, F. Huang, C. Yang, F. Noblesse, Errors due

- to a practical Green function for steady ship waves, *European Journal of Mechanics-B/Fluids* 55 (2016) 162-169.
- [23] F. Noblesse, G. Delhommeau, F. Huang, C. Yang, Practical mathematical representation of the flow due to a distribution of sources on a steadily advancing ship hull, *Journal of Engineering Mathematics* 71 (4) (2011) 367-392.
- [24] H. Wu, C. Ma, Y. Zhu, Z. Yang, W. Li, F. Noblesse, Approximations of the Green function for diffraction radiation of water waves, *The 26th International Ocean and Polar Engineering Conference*, Rhodes, Greece, (2016) 134-139.
- [25] F. Noblesse, Integral identities of potential theory of radiation and diffraction of regular water waves by a body. *Journal of Engineering Mathematics* 17 (1) (1983) 1-13.
- [26] A.J.M. Hitchcock, Polynomial approximations to Bessel functions of order zero and one and to related functions, *Mathematics of Computation* 11 (58) (1957) 86-88.
- [27] M. Abramowitz, I.A. Stegun, *Handbook of mathematical functions*, Dover, New York (1965).
- [28] Y.L. Luke, *Mathematical functions and their approximations*, Academic Press, New York (1975).
- [29] J.N. Newman, Approximations for the Bessel and Struve functions, *Mathematics of Computation* 43 (168) (1984) 551-556.

'A global approximation to the Green function for diffraction radiation of water waves'

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Highlights

1. The Green function of the theory of diffraction radiation of time-harmonic (regular) waves by an offshore structure, or a ship at low speed, in deep water is considered.
2. The Green function G and its gradient are expressed in the usual manner as the sum of three components that correspond to the fundamental free-space singularity, a non-oscillatory local flow, and waves.
3. Simple approximations that only involve elementary continuous functions (algebraic, exponential, logarithmic) of real arguments are given for the local flow components in G and its gradient.
4. These approximations are **global approximations valid within the entire flow region**, rather than within complementary contiguous regions as can be found in the literature.
5. The analysis of the errors associated with the approximations to the local flow components given in the study shows that the approximations are sufficiently accurate for practical purposes.
6. These global approximations provide a particularly simple and highly efficient way of numerically evaluating the Green function and its gradient for diffraction radiation of time-harmonic waves in deep water.